



Agreement and bias between direct LDL-C measurement and estimation equations in routine clinical lipid samples at Sebbi Hospital, Uganda

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ABSTRACT

Background: Low-density lipoprotein cholesterol (LDL-C) is central to cardiovascular risk assessment, yet its accurate determination remains challenging. The Friedewald formula performs poorly in hypertriglyceridemia, while newer formulas such as Martin/Hopkins and Sampson show improved accuracy. In Uganda and sub-Saharan Africa, contemporary comparisons of these equations against direct assays are lacking, increasing the risk of misclassification.

Methods: We conducted a retrospective method comparison study at Sebbi Hospital, Uganda, analyzing 1289 lipid profiles (2022–2025). Directly measured LDL-C on the Mindray BS-230 analyzer was compared with Friedewald, Martin/Hopkins, and Sampson formula-calculated LDL-C. Agreement, correlation, root mean square error (RMSE), mean absolute difference (MAD), and misclassification at European Society of Cardiology/European Atherosclerosis Society (ESC/EAS) treatment thresholds were assessed across triglyceride strata (<150, 150–399, ≥400 mg/dL).

Results: For TG <150 mg/dL, all methods showed comparable agreement with the Sampson equation (84.13 – 85.03%, $k = 0.696 - 0.706$), with misclassification rates of 14.97 – 15.87% and MAD ranging from 13.87 to 15.54 mg/dL. For TG 150 – 399 mg/dL, agreement declined markedly for Friedewald (67.75%), but remained substantially higher for directly measured LDL-C (86.07%) and Martin/Hopkins (82.12%). Misclassification was highest for Friedewald (32.43%) compared with Martin/Hopkins (17.88%) and directly measured LDL-C (13.93%). For TG ≥400 mg/dL, agreement was nominally high for directly measured LDL-C (78.79) but weighted kappa was poor and non-significant for all three methods. MAD was 57.56mg/dL for directly measured LDL-C,

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103.45 mg/dL for Friedewald and 64.96 mg/dL for Martin/Hopkins, with over 93% of directly measured LDL-C values differing from Sampson-calculated values by more than 20 mg/dL.

Conclusion: Agreement and misclassification varied by triglycerides level, with all methods showing comparable concordance with Sampson equation in the low-TG stratum. Friedewald equation showed the poorest agreement and highest misclassification at moderately elevated triglycerides level. In severe hypertriglyceridemia, large differences were observed across all methods, though conclusions were limited by small sample size. Findings highlight the importance of transparent reporting of LDL-C measurement method and careful interpretation near ESC/EAS treatment thresholds.

Introduction

Low-density lipoprotein cholesterol (LDL-C) remains the primary biomarker for cardiovascular risk assessment and the principal target of lipid-lowering therapy [1,2]. Accurate LDL-C determination is critical for clinical decision-making, given that elevated LDL-C levels are consistently associated with increased risk of atherosclerotic cardiovascular disease (ASCVD) events, including myocardial infarction and stroke [3].

Traditionally, LDL-C was estimated using the Friedewald equation, which calculates LDL-C indirectly from measurements of total cholesterol (TC), high-density lipoprotein cholesterol (HDL-C), and triglycerides (TG) [4]. However, this formula has well-documented limitations, particularly in patients with hypertriglyceridemia (TG > 400 mg/dL) or very low LDL-C values (<70 mg/dL), where it systematically underestimates or overestimates LDL-C values respectively [5,6]. This is due to the fact that the $\frac{TC}{TG}$ ratio in very low density lipoproteins (VLDL) vary considerably, and the Friedewald formula requires fasting serum to accurately estimate LDL-C since chylomicronemia in a non-fasting situation leads to the overestimation of VLDL-C [1,7]. To address these limitations, newer calculation methods, notably the Martin/Hopkins and Sampson equations, have been validated across diverse patient populations, demonstrating improved performance in challenging lipid profiles. The Martin/Hopkins equation is more accurate than the Friedewald equation, particularly for low-LDL-C samples [8,9] that are now more common because of more effective lipid-lowering therapy, with the Sampson equation demonstrating further improvements in accuracy [5,9,10].

In parallel, direct LDL-C assays have gained widespread adoption in clinical laboratories due to their automation, convenience, and independence from fasting status. These homogeneous assays are intended to improve accuracy by directly quantifying LDL-C without reliance on calculation-based algorithms [11,12]. However, recent large-scale evaluations have revealed significant concerns about their performance. In a multi-platform study, Meeusen et al., (2025) found that recently validated calculation formulas consistently outperformed direct LDL-C assays, with the modified Sampson equation showing the best overall correlation with the reference beta-quantification method, especially in patients with hypertriglyceridemia (TG > 400 mg/dL) or LDL-C < 70 mg/dL [13]. They concluded that in patients with this profile, calculated LDL-C should replace direct measurement assays when guiding clinical decisions. Similar findings had been previously reported in high-triglyceride populations [14,15]. Additionally, guideline endorsement of non-fasting samples for initial lipid screening has raised concerns about the accuracy of directly measured LDL-C, particularly in the presence of markedly elevated triglycerides (>400 mg/dL) [3].

The clinical implications of LDL-C measurement discrepancies are substantial. The recent 2025 European Society of Cardiology/European Atherosclerosis Society (ESC/EAS) focused update of the 2019 guidelines for the management of dyslipidemias defines the recommended LDL-C treatment goals as <40 mg/dL (1.0 mmol/L) for extreme risk patients, <55 mg/dL (<1.4 mmol/L) for very high-risk patients, <70 mg/dL (<1.8 mmol/L) for high-risk patients, and <100 mg/dL (<2.6 mmol/L) for moderate risk patients [16]. Therefore, misclassification may lead to inappropriate treatment decisions, either by underestimating patients' cardiovascular risk or by exposing others to unnecessary therapy, drug side effects, and associated costs.

In sub-Saharan Africa, cardiovascular diseases (CVDs) represent a growing public health challenge. Out of the 17 million premature deaths (under the age of 70) due to non-communicable diseases that occurred in 2015, 82 % occurred in low-and middle-income countries, and 37 % were caused by CVDs [17]. In Uganda, CVDs account for 10 % of premature deaths and disability (for adults 30 – 70years), with an age-standardized rate of 9.85 % [18].

Across health facilities in Uganda and many laboratories across sub-Saharan Africa, lipid profile testing is performed using a wide range of automated and semi-automated chemistry analyzers, with LDL-C assays inclusive. An earlier study at Makerere university that compared direct LDL-C assays with Friedewald calculation among fasting students found incongruities between the methods, noting that clinicians were often unaware of which method their laboratory used [19]. Fifteen years later, no contemporary study has compared newly validated equations like the Martin/Hopkins and Sampson equations, against direct LDL-C assays in routine clinical practice in Uganda especially for patients with high TG (> 400 mg/dl).

Without this local relevant data on the comparative performance of LDL-C measurement methods, particularly in patients with hypertriglyceridemia or very low LDL-C, there is risk of systemic misclassification of cardiovascular risk and inconsistent care across facilities. While beta quantification remains the gold standard for LDL-C determination, it is not feasible in our routine resource limited clinical settings. However, Sampson's equation has demonstrated the closest agreement with beta quantification across the full range of triglycerides and LDL-C values in validation study [13], making it a suitable comparator for method agreement assessments. We aimed to compare directly measured LDL-C values obtained by the Mindray BS 230 biochemistry analyzer with those calculated using Friedewald, Martin/Hopkins and Sampson equations in routine clinical samples at Sebbi Hospital, Uganda, using the Sampson's

equation as the primary comparator. By generating locally relevant evidence on the agreement between direct assays and calculated LDL-C methods in routine practice, this study aims to inform laboratory protocols and clinical decision-making, supporting more reliable cardiovascular risk assessment and treatment optimization across Ugandan health facilities

Methods

Study design

This retrospective method comparison study was conducted at Sebbi Hospital, Uganda using routinely generated laboratory data. Data was collected from lipid profile tests conducted between January 1, 2022 and December 31st, 2025. All lipid profile results were extracted in the analyzer-reported units (mmol/L) and converted to mg/dL for analysis and reporting

Study population

The study population consisted of all patients who underwent lipid profile testing as part of routine clinical care during the study period. Lipid testing was requested by clinicians for cardiovascular risk assessment, treatment monitoring, or routine health screening.

Eligibility criteria

Inclusion criteria

- Complete lipid panel with all four components measured: TC, TG, HDL-C, LDL-C
- Results available in the hospital information system (HIS) with no missing values

Exclusion criteria

- Incomplete lipid panels (Missing one or more analytes)
- Samples with documented pre-analytical issues (hemolysis, icterus, or lipemia flags) as documented in the HIS

Sample size determination

No formal sample size calculation was performed a priori. All eligible lipid profiles available during the four-year study period were included, yielding 1289 eligible lipid profiles. Post hoc power was assessed within the paired t -test framework, assuming a two-sided α of 0.05 and a conservative standard deviation of paired differences ($\delta_d = 10 \text{ mg/dL}$) to avoid overestimation of statistical power. Under these assumptions, power depended on the standardized mean difference $\mu_d / \left(\frac{\delta_d}{\sqrt{n}} \right)$, where μ_d is the mean LDL-C difference, δ_d is the standard deviation of paired differences, and n is the number of paired observations. Given $n = 1289$, the study had $> 90\%$ statistical power to detect a mean difference of 1 mg/dL and $> 99\%$ statistical power to detect a mean difference of 2 mg/dL. Including all eligible lipid profiles maximized precision for agreement and misclassification analyses

Sampling stratification

To allow subgroup representation, all lipid profile tests were categorized according to clinically important strata

- Triglycerides < 150 , $150 \leq \text{Triglycerides} < 400 \text{ mg/dL}$ and Triglycerides $\geq 400 \text{ mg/dL}$
- Direct LDL-C stratified by the treatment thresholds of $< 40 \text{ mg/dL}$, $40\text{--}55 \text{ mg/dL}$, $55\text{--}70 \text{ mg/dL}$, $\geq 70 \text{ mg/dL}$.

This stratification permitted focused assessment of agreement and bias within clinically important lipid phenotypes while retaining full dataset for overall method comparison: direct LDL-C vs Friedewald, Martin/Hopkins, and Sampson equations.

Data sources and variables

Lipid profile data were extracted from HIS called Streamline. For each eligible test, the following were retrieved: Laboratory number, patient age (years), patient sex (male/ female), date of sample collection, date of testing, TC, TG, HDL-C, LDL-C. Fasting status was not recorded to reflect routine clinical practice where both fasting and non-fasting samples are analyzed. Laboratory numbers were, after data extraction accuracy checks by a second reviewer, replaced with unique study numbers before analysis. While multiple lipid profiles were available for the same patient during the study period (identified using the unique hospital numbers), each test was treated as an independent observation.

Laboratory methods

All measurements were performed using the Mindray BS 230 automated chemistry analyzer using manufacturer-provided reagents and calibrators. The Mindray BS-230 is an automatic, discrete, random-access benchtop analyzer capable of up to 200 tests/hour (400 with ISE), using absorbance photometry, turbidimetry, and optional ion-selective electrodes with flexible reagent and sample handling. It features 80 reagent and 40 sample positions with automatic dilution, cuvette-based reactions, and precise temperature control (37 ± 0.1 °C). The system includes multi-wavelength optics, various calibration/control options, optional barcoding and Laboratory information system (LIS) connectivity [20].

Index test: LDL-C principle of measurement

In the first step of the reaction, the LDL is selectively protected while HDL, VLDL, Chylomicrons (CM) are treated by enzymes. In the second step, LDL is released, and LDL-C is selectively measured by the cholesterol enzymatic chromogenic reaction. LDL-C is converted to cholestenone and H_2O_2 by cholesterol esterase (CHE) and cholesterol oxidase (CHO). Catalyzed by peroxidase (POD), the H_2O_2 reacts with 4-aminophenazone and N-ethyl-N-(2-hydroxy-3-sulfopropyl)-m-toluidine sodium (TOOS) to form quinonimine chromogen. The absorbency increase is directly proportional to the concentration of LDL-C.

The test is measured in mmol/L, has a measurement range of 0.05 – 20 mmol/L (conversion factor: $mg/dL \times 0.0259 = mmol/L$).

Input variables

Triglycerides (TG)

Mindray BS 230 measures TG using Glycerokinase Peroxidase-Peroxidase (POD-POD) methods. Through a sequence of enzymatic catalysis by lipase, Glycerokinase (GK) and Glycerol-3-phosphate-oxidase (GPO), triglycerides is catalyzed to yield H_2O_2 . Catalyzed by Peroxidase, the H_2O_2 reacts with 4-aminophenazone and DHBS to yield a colored dye of quinonimine. The absorbency increase is directly proportional to the concentration of triglycerides.

The test is measured in mmol/L, has a measurement range of 0.1 – 12.5 mmol/L (conversion factor: $mg/dL \times 0.0113 = mmol/L$).

Total cholesterol

This is measured using the Cholesterol oxidase – peroxidase (CHOD – POD) method. Cholesterol esters are cleaved by the action of cholesterol esterase (CHE) to yield free cholesterol and fatty acids. The free cholesterol is then oxidized with the aid of cholesterol oxidase (CHO) and consumption of oxygen, giving rise to H_2O_2 . Catalyzed by Peroxidase (POD), the H_2O_2 reacts with 4-aminophenazone and TOOS to form a quinonimine chromogen. The absorbency increase is directly proportional to the concentration of cholesterol.

The test is measured in mmol/L, has a measurement range of 0.1 – 20.0 mmol/L (conversion factor: $mg/dL \times 0.0259 = mmol/L$).

HDL-C

HDL-C is measured using a direct PEG-modified enzyme/sulfate α -cyclodextrin method. In the first step of the reaction, LDL, VLDL, Chylomicrons (CM) conjugate with sulfate α -cyclodextrin, forming polymers in the presence of Mg^{2+} . In the second step, HDL-C is converted to cholestenone and H_2O_2 by PEG-modified cholesterol esterase (CHE) and cholesterol oxidase (CHO). Catalyzed by Peroxidase (POD), the H_2O_2 reacts with 4-aminophenazone and N-ethyl-N-(2-hydroxy-3-sulfopropyl)-m-toluidine sodium (TOOS) to form quinonimine chromogen. The absorbency increase is directly proportional to the concentration of HDL-C.

The test is measured in mmol/L, has a measurement range of 0.05 – 6.0 mmol/L (conversion factor: $mg/dL \times 0.0259 = mmol/L$).

Comparator method

Sampson's equation was used as the primary comparator across all triglycerides strata. This choice was based on: (i) its validation against beta quantification [13], (ii) demonstrated superior performance compared to Friedewald and Martin/Hopkins equations [5, 13], and (iii) the absence of beta quantification in our routine clinical setting. Agreement between directly measured LDL-C (Mindray BS 230), Friedewald-calculated LDL-C, and Martin/Hopkins-calculated LDL-C was assessed relative to Sampson-calculated values.

Friedewald equation

The Friedewald equation was derived from a sample of 448 individuals with familial hyperlipoproteinemia or their relatives in 1972 [4]. The equation

$$LDL - C = TC - HDL - C - \frac{TG}{5}.$$

A fixed factor of 5 was used to describe the relationship between TG and VLDL-C (TG: VLDL). However, the actual TG:VLDL-C ratio varies significantly across the range of triglyceride and cholesterol levels [9].

Martin/Hopkins equation

This equation was derived from a sample of 1.35 million children, adolescents, and adults in the United States, with lipid distributions closely matching the population-based National Health and Nutrition Examination Survey (NHANES) [9].

$$LDL = TC - HDL - \frac{TG}{\text{novel factor}}$$

Where the *novel factor* is an adjustable factor based on an individual's non-HDL and TG level and could be obtained from a 6 by 30 table. The equation improves the LDL-C estimation accuracy at clinically relevant low LDL-C and moderately elevated TG (150 – 400 mg/dL) and has been validated by several studies [1,21,22]

LDL-C was calculated by applying the individualized TG:VLDL-C factor using the published 180-cell Excel calculator whose inputs are each patient's TC, HDL-C, and TG in mg/dL [23].

Sampson equation

Developed in 2020, the Sampson equation was derived from results of patients with TG levels as high as 9 mmol/L or 800 mg/dL [5]. It was based beta- quantification

$$LDL = \frac{TC}{0.948} - \frac{HDL}{0.971} - \left(\frac{TG}{8.56} - \frac{TG \times (TC - HDL)}{2140} - \frac{TG^2}{16100} \right) - 9.44$$

Samples preparation

For all tests TG, TC, HDL-C and LDL-C, serum samples were used. Samples were collected in *Equi Vac clot activator tubes*, allowed 20 - 30 min of clotting, and centrifuged using the Hamilton Bell Vanguard V6500 centrifuge that operates at a fixed maximum speed of approximately 3400 rpm (± 100 rpm) under normal load conditions, producing a maximum relative centrifugal force (RCF) of about $1318 \times g$ with its six-place fixed-angle rotor [24].

Statistical analysis

Descriptive statistics

Age was summarized as median (range, in years). Categorical variables (TG treatment categories) were described using median (range), skewness and kurtosis. The distribution of samples across LDL -C treatment categories (<40, 40 – 54, 55 – 69, 70 – 99, >100 mg/dL) was reported as frequencies.

Method comparison

Method agreement was defined as the proportion of samples showing concordant treatment category assignment between each method (directly measured LDL-C (dLDL-c), Friedewald, Martin/Hopkins) and the Sampson's equation, stratified by ESC/EAS LDL-C treatment thresholds: <40 mg/dL, <55 mg/dL, <70 mg/dL, and <100 mg/dL. Cohen's kappa, correlation coefficient (r), root mean square error [RMSE] were calculated for each comparison. The magnitude of difference was defined as Sampson-calculated LDL-C minus the comparison method (directly measured LDL-C (Mindray BS 230), Friedewald, and Martin/Hopkins-calculated LDL-C) across all triglyceride strata. Differences were further summarized as the percentages of LDL-C results with absolute differences of <6.6 mg/dl, 6.6 – 20 mg/dl, and >20 mg/dl (based on CLIA TEa = $\pm 20\%$, Bias = 0.33Tea) [25]. We calculated the mean absolute difference (MAD) between the Sampson equation and each comparison method. MAD values were assessed using a general linear model (GLM) with repeated measures and further evaluated using the non-parametric Wilcoxon Signed Rank test, with residual error blots used for visualization. We computed the percentage of samples showing discordant treatment category assignment (misclassification) between each method and the Sampson equation at LDL-C treatment thresholds of <40 mg/dL, 40-<55 mg/dL, 55-<70 mg/dL, ≥ 70 mg/dL. All analyses were performed using R software (version 4.5.2) with statistical significance set at $\alpha = 0.05$. Study limitation and interpretation framework.

Beta quantification, the reference method for LDL-C determination is not available in routine clinical care in Uganda and many resource-limited settings. Therefore, this study represents a method comparison rather than an accuracy validation study. The Sampson equation was selected as the primary comparator based on its demonstrated closest agreement with beta quantification in validation studies [5,13]. Our findings therefore reflect concordance in treatment category assignment and agreement between methods rather than absolute analytical accuracy. Discordant results between methods indicate potential clinical decision-making implications but not definitively establish which method is "correct" in the absence of beta-quantification. Additionally, repeated lipid profiles from the same patients were treated as independent observations, which may affect precision estimates. However, this approach reflected the real-world clinical practice where each test informs clinical decisions independently.

Results

Characteristics of study patients

A total of 1289 lipid profiles were analyzed. Based on triglycerides levels, 775 samples (60.12 %) had $TG < 150$ mg/dL, 481 samples (37.32 %) had $150 \leq TG < 400$ mg/dL, and 33 samples (2.56 %) had $TG \geq 400$. The baseline characteristics and lipid profile distributions across triglycerides strata are summarized in Table 1.

Samples with $TG < 150$ mg/dL ($n = 775$)

Among the 775 samples with $TG < 150$ mg/dL, median age was 48 yrs (range 12 – 106), median TC 185.33 mg/dL, median TG 95.58 mg/dL, median HDL-C was 50.97 mg/dL, and median dLDL-C was 120.46 mg/dL. Agreement with Sampson equation was high across all methods: 84.13 % ($k = 0.696$, $p < 0.001$) for directly measured LDL-C, 84.26 % ($k = 0.701$, $p < 0.001$) for Friedewald, and 85.03 % ($k = 0.706$, $p < 0.001$) for Martin/Hopkins. Spearman correlations with Sampson's equation were 0.879 for directly measured LDL-C, 0.992 for Friedewald, and 0.997 for Martin/Hopkins (Fig. 1a-c).

RMSE was lowest for Friedewald (15.42 mg/dL) and Martin/Hopkins (15.51 mg/dL), with directly measured LDL-C showing higher values (23.30 mg/dL). MAD ranged from 13.87 mg/dL for Friedewald to 15.54 mg/dL for directly measured LDL-C, with statistically significant differences across methods by GLM repeated measures ($F = 4.71$, $p = 0.009$, Fig. 2a-c). All three methods differed significantly from Sampson equation by pairwise Wilcoxon signed-rank test ($p < 0.001$ for all).

Misclassification rates relative to Sampson-defined treatment thresholds were 15.87 % for directly measured LDL-C, 15.74 % for Friedewald, and 14.97 % for the Martin/Hopkins equation. Directly measured LDL-C showed a higher proportion of samples with differences exceeding 20 mg/dL (27.0 %) compared with Friedewald (19.4 %) and Martin/Hopkins (17.9 %) (Table 2).

Samples with $150 \leq TG < 400$ mg/dL

Among the 481 samples with moderately elevated triglycerides, median age was 48 years (range 9 – 95), median TC 208.88 mg/dL, median TG was 209.73 mg/dL, median HDL-C was 44.02 mg/dL, median dLDL-C was 132.82 mg/dL and median Sampson-calculated LDL-C 155.05 mg/dL.

Agreement with the Sampson equation declined compared with the lower TG stratum. Directly measured LDL-C showed 86.07 % agreement ($k = 0.527$, $p < 0.001$), while Friedewald showed 67.57 % agreement ($k = 0.166$, $p < 0.001$), and Martin/Hopkins 82.12 % agreement ($k = 0.490$, $p < 0.001$). Spearman correlations with Sampson's equation were 0.883 for directly measured LDL-C, 0.956 for Friedewald, and 0.985 for Martin/Hopkins equations. (Fig. 1d-f). RMSE was lowest for Martin/Hopkins (34.18 g/dL), followed by directly measured LDL-C (35.45 mg/dL), and highest for Friedewald (43.68 mg/dL). MAD followed the similar pattern: Martin/Hopkins 31.28 mg/dL, directly measured LDL-C 26.89 mg/dL and Friedewald 40.82 mg/dL. GLM repeated measures confirmed significant differences across methods ($F = 75.63$, $p < 0.001$), with all pairwise Wilcoxon signed-rank comparisons significant ($p < 0.001$). Residual error plots revealed systematic directional bias, with Friedewald equation markedly underestimating LDL-C relative to Sampson's equation (Fig. 2d-f). Misclassification was highest for Friedewald (32.43 %) compared with Martin/Hopkins (17.88 %) and directly measured LDL-C (13.93 %).

Table 1

Classification concordance of LDL-C measurement methods relative to Sampson equation, stratified by triglyceride level.

TG Stratum	LDL-C Threshold (mg/dL)	Sampson n	Direct LDL-C n (%)	Friedewald n (%)	Martin/Hopkins n (%)
TG <150 mg/dL (N = 775)	<40	10	6 (60.0 %)	10 (100.0 %)	10 (100.0 %)
	40 -54	24	12 (50.0 %)	18 (75.0 %)	17 (70.8 %)
	55 -69	30	19 (63.3 %)	18 (60.0 %)	23 (76.7 %)
	70 - 99	144	109 (75.7 %)	116 (80.6 %)	118 (81.9 %)
	≥ 100	567	506 (89.2 %)	491 (86.6 %)	491 (86.6 %)
TG 150 - 399 mg/dL (N = 481)	<40	0	-	-	-
	40 - 54	0	-	-	-
	55 - 69	8	3 (37.5 %)	0 (0.0 %)	1 (12.5 %)
	70 - 99	46	25 (54.3 %)	9 (19.6 %)	29 (63.0 %)
	≥ 100	427	386 (90.4 %)	316 (74.0 %)	365 (85.5 %)
TG ≥ 400 mg/dL (N = 33)	<40	0	-	-	-
	40 - 54	0	-	-	-
	55 - 69	0	-	-	-
	70 - 99	1	1 (100.0 %)	0 (0.0 %)	0 (0.0 %)
	≥ 100	32	25 (78.1 %)	9 (28.1 %)	21 (65.6 %)

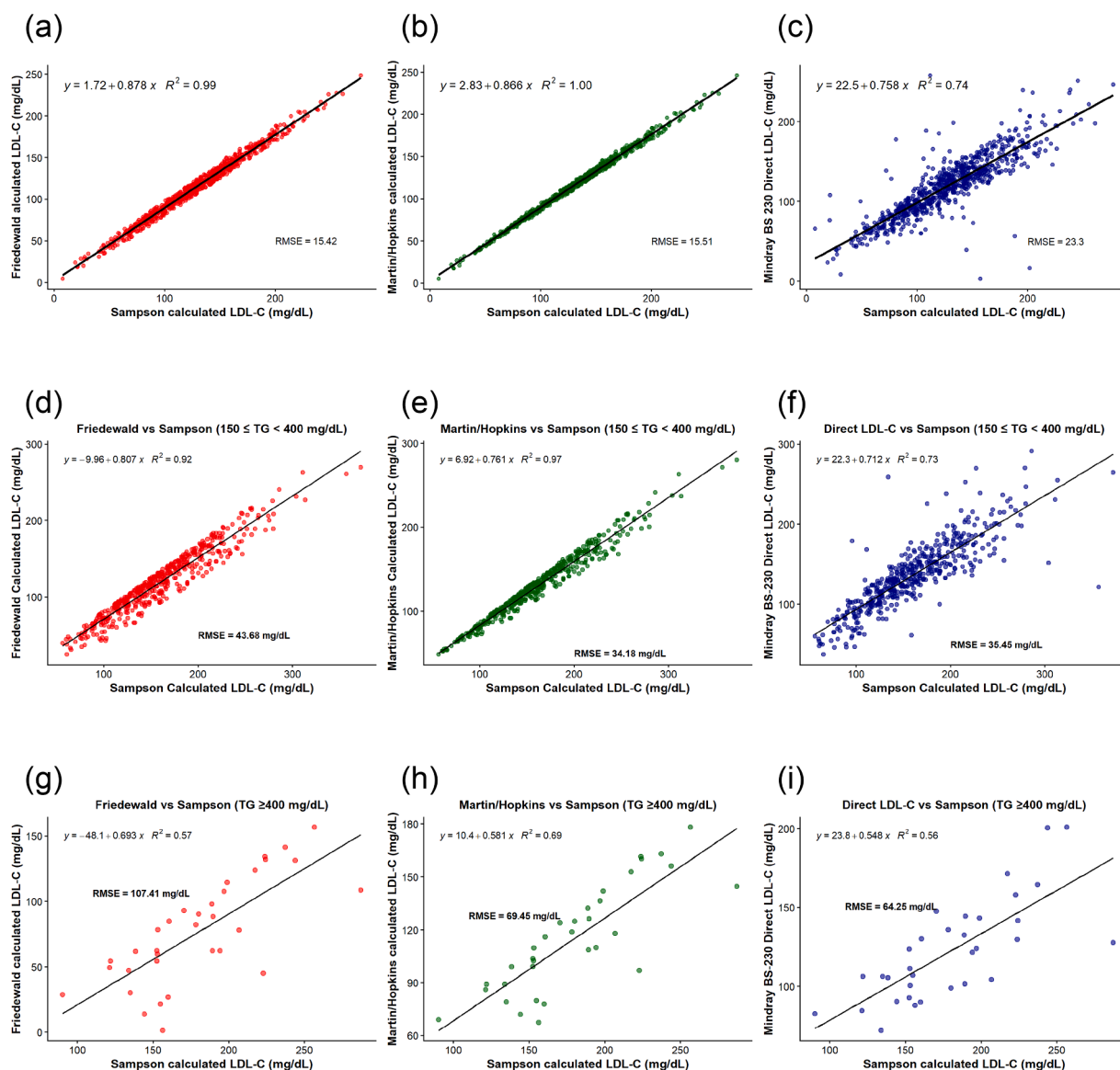


Fig. 1. Linear regression analyses comparing LDL-C values from three methods against Sampson-calculated LDL-C across triglyceride strata. Panels (a–c) present comparisons of Friedewald, Martin/Hopkins, and directly measured LDL-C (Mindray BS-230), respectively, against Sampson-calculated LDL-C for samples with $TG < 150$ mg/dL ($n = 775$). Panels (d–f) present corresponding comparisons for samples with $TG 150–399$ mg/dL ($n = 481$). Panels (g–i) present comparisons for samples with $TG \geq 400$ mg/dL ($n = 33$). The Sampson equation serves as the primary comparator across all strata. The solid line represents the fitted regression line and R^2 values are shown for each panel.

Samples with $TG \geq 400$ mg/dL

For the 33 samples with severe hypertriglyceridemia, the median age was 43 years (range 32 – 64), median TC 213.90 mg/dL, median TG 479.65 mg/dL, median HDL-C 35.14 mg/dL, and median dLDL-C was 121.62 mg/dL. As Friedewald and Martin/Hopkins formulas are not validated at this triglycerides level, all three methods were for exploratory purposes, compared with Sampson's equation. Agreement with Sampson's equation was 78.78 % (95 % CI 61.09 – 91.02 %) for directly measured LDL-C, 27.27 % (95 % CI 13.30 – 45.52 %) for Friedewald equation and 63.64 % (95 % CI 45.12 – 79.60 %) for Martin/Hopkins. Weighted kappa was low for all the three methods: directly measured LDL-C ($k = 0.178$, $p = 0.073$), Friedewald equation ($k = -0.113$, $p = 0.163$), and Martin/Hopkins ($k = 0.031$, $p = 0.710$), likely reflecting the limited power from the small sample size in this stratum. Spearman correlations were 0.745 for directly measured LDL-C, 0.771 for Friedewald equation, and 0.833 for Martin/Hopkins equation. MAD was 57.56 mg/dL for directly measured LDL-C, 103.45 mg/dL for Friedewald equation, and 64.96 mg/dL for Martin/Hopkins formula (Fig. 2g-i). Differences exceeding 20 mg/dL were observed in 93.94 % of directly measured LDL-C values and 100 % of both Friedewald and Martin/Hopkins-calculated values. Misclassification rates were 21.21 % for directly measured LDL-C, 72.73 % for Friedewald equation, and 36.36 % for

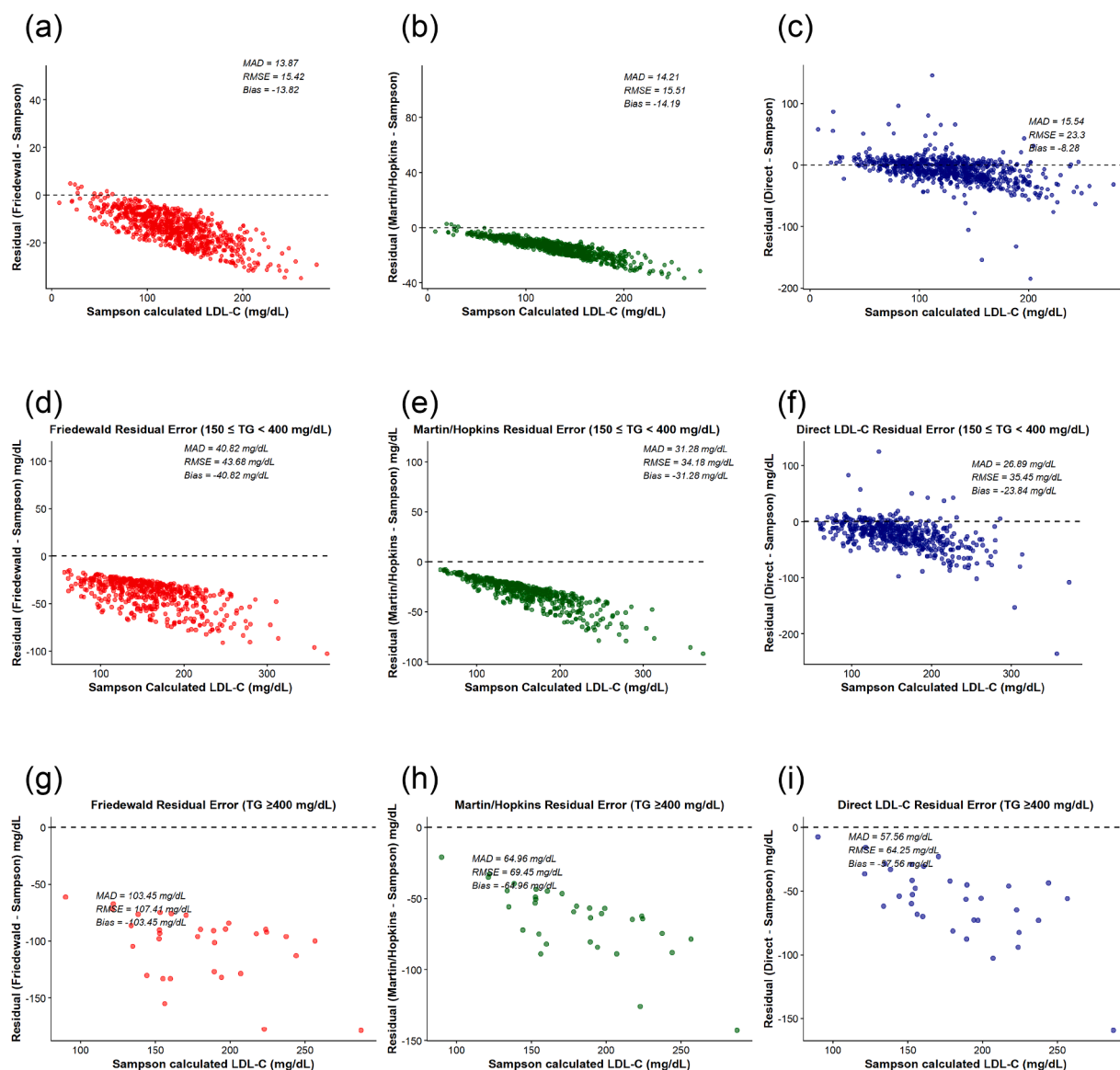


Fig. 2. Residual difference plots showing differences between each method and Sampson-calculated LDL-C across triglyceride strata. Panels (a–c) display residuals for Friedewald, Martin/Hopkins, and directly measured LDL-C (Mindray BS-230), respectively, for samples with TG <150 mg/dL ($n = 775$). Panels (d–f) show corresponding residuals for samples with TG 150–399 mg/dL ($n = 481$). Panels (g–i) present residuals for samples with TG ≥400 mg/dL ($n = 33$). The horizontal dashed line at zero indicates perfect agreement with Sampson; negative values indicate underestimation and positive values indicate overestimation relative to Sampson-calculated LDL-C. MAD = mean absolute difference; RMSE = root mean square error; Bias = mean difference.

Martin/Hopkins equation.

Discussion

This study evaluated agreement, correlation, and misclassification between directly measured LDL-C, Friedewald, and Martin/Hopkins equations against the Sampson's equation as the primary comparator, across clinical triglyceride strata in routine Ugandan practice. In the absence of beta quantification in our routine clinical settings, misclassification was defined as disagreement in LDL-C treatment category (<40, <55, <70, <100 mg/dL) relative to Sampson-defined categories within each stratum. The Sampson's equation was selected as the primary comparator across all triglyceride's strata based on its demonstrated closest agreement with beta quantification in validation studies [5,13]. Discordant results between methods therefore indicate potential clinical decision-making implications but do not definitely establish which method is correct in the absence of beta quantification.

In the low-TG stratum (TG <150 mg/dL), all three methods showed comparable agreement with the Sampson equation, exceeding

Table 2

Agreement and difference metrics of LDL-C measurement methods relative to Sampson equation, stratified by triglyceride level.

TG Stratum	Method	Agreement % (95 % CI)	Weighted Kappa	Spearman r	RMSE (mg/dL)	MAD (mg/dL)	<6.6 mg/dL %	6.6 -20 mg/dL %	>20 mg/dL %	Misclassification %
TG <150 mg/ dL (N = 775)	Direct LDL- C	84.13 (81.36 - 86.63)	0.696	0.879	23.30	15.54	32.5	40.5	27.0	15.87
	Friedewald	84.26 (81.50 -86.75)	0.701	0.992	15.42	13.87	15.5	65.2	19.4	15.74
	Martin/ Hopkins	85.03 (82.32 -87.47)	0.706	0.997	15.51	14.21	10.3	71.7	17.9	14.97
TG 150–399 mg/dL (N = 481)	Direct LDL- C	86.07 (82.65 -89.04)	0.527	0.883	35.45	26.89	14.6	31.6	53.8	13.93
	Friedewald	67.57 (63.18 -71.74)	0.166	0.956	43.68	40.82	0.0	2.3	97.7	32.43
	Martin/ Hopkins	82.12 (78.40 -85.44)	0.490	0.985	34.18	31.28	0.0	19.3	80.7	17.88
TG ≥400 mg/ dL (N = 33)	Direct LDL- C	78.79 (61.09 -91.02)	0.178	0.745	64.25	57.56	0.0	6.1	93.9	21.21
	Friedewald	27.27 (13.30 -45.52)	-0.113	0.771	107.41	103.45	0.0	0.0	100.0	72.73
	Martin/ Hopkins	63.64 (45.12 -79.60)	0.031	0.833	69.45	64.96	0.0	0.0	100.0	36.36

84 %, with similar misclassification rates ranging from 14.97 % to 15.87 %. It is important to note that although Spearman's correlations were strong, correlation does not imply agreement, as two methods can correlate strongly while showing systematic differences. The findings are consistent with prior validation studies demonstrating that formula-based differences are minimized when triglycerides are within the normal clinical range, [1,22,26]. Notably, Cicero et al., (2021) similarly demonstrated in a large cohort of children that Sampson's equation was more reliable than Friedewald's even at extreme triglycerides values, when compared against directly measured LDL-C [27]. Together, these findings suggest that in the low-TG stratum, method choice is less likely to affect clinical decision-making near ESC/EAS treatment thresholds.

Performance diverged substantially in the moderately elevated TG stratum ($150 \leq \text{TG} < 400$ mg/dL). Agreement with the Sampson equation fell markedly for Friedewald (67.75, $k = 0.166$) but remained substantially higher for directly measured LDL-C (86.07 %, $k = 0.527$) and Martin/Hopkins (82.12 %, $k = 0.490$). The systematic directional bias visible in the residual plots, is consistent with well-recognized limitation of fixed TG:VLDL-C ratio assumed by the Friedewald equation at elevated triglycerides levels [4]. The findings further align with previously reported improved concordance of Martin/Hopkins and Sampson's equations over Friedewald in moderately elevated TG populations [1,2]. In a large Italian population study of 114,774 individuals, Piani et al., (2021) similarly found that Sampson's and Martin/Hopkins equations were the most accurate, while Friedewald showed the highest misclassification, particularly at elevated triglycerides levels [28]. Clinically, patients near ESC/EAS treatment thresholds in this stratum face a 32.43 % risk of mis-categorization when Friedewald is used, potentially leading to undertreatment if LDL-C is underestimated or unnecessary therapy intensification if overestimated [15].

In the severe hypertriglyceridemia stratum ($\text{TG} \geq 400$ mg/dL), agreement was nominally highest for directly measured LDL-C (78.79 %), but with a poor weighted kappa ($k = 0.178$, $p = 0.073$), indicating that apparent agreement may reflect chance concordance rather than true method agreement. Friedewald and Martin/Hopkins showed even lower agreement (27.27 % and 63.64 % respectively) with kappa values that were poor and non-significant for both. This stratum had the largest differences across all methods, with MAD reaching 57.56 mg/dL for directly measured LDL-C, 103.45 mg/dL for Friedewald, and 64.96 mg/dL for Martin/Hopkins equation. These findings reflected the concerns raised by Meeusen et al. (2025) [13] about direct assay performance in severe hypertriglyceridemia. However, given the small sample size ($n = 33$), these findings are only exploratory, and do not definitively establish the superiority of any method in this stratum. The observed pattern of large differences, with over 93 % of directly measured LDL-C values differing from the Sampson-calculated values by more than 20 mg/dL, nevertheless highlights the clinical risk of relying on any single method near ESC/EAS treatment thresholds in patients with severe hypertriglyceridemia. Larger studies with beta quantification as the reference are needed to confirm these findings in Ugandan populations.

These findings provide locally relevant evidence on the comparative performance of directly measured LDL-C and three calculation equations across triglycerides strata in routine Ugandan clinical practice. Agreement and misclassification varied by triglyceride level, with the greatest divergence observed in the moderate hypertriglyceridemia stratum, where Friedewald calculated values showed the

poorest concordance with Sampson equation-calculated values. In severe hypertriglyceridemia, larger differences between all methods and Sampson-calculated LDL-C highlight the importance of cautious interpretation of results in this range, though conclusions remain limited by the small sample size in this stratum. The broader relevance of equation-based estimation of lipid biomarkers in resource limited settings is further supported by Fogacci et al., (2025), who demonstrated that simple, low-cost equation-based approaches can reliably estimate ApoB levels in settings where direct measurement is unavailable or expensive [29], reinforcing the case for validated equation-based LDL-C estimation in similar resource constrained environments such as Uganda. Taken together, our findings support the case for transparent reporting of the LDL-C measurement method used across health facilities, and for clinicians to interpret LDL-C values in the context of the patient's triglyceride level, particularly when results fall near the ESC/EAS treatment thresholds.

Limitations

This study has several limitations that should be considered when interpreting its findings. Beta quantification, the reference method for LDL-C determination, was not available, and therefore this study represents a method comparison rather than an accuracy validation. Discordant results between methods indicate potential clinical implications but do not definitively establish which method is correct. Fasting status was not recorded, reflecting routine clinical practice but limiting direct comparability with fasting-based validation studies. All measurements were performed on a single analyzer platform (Mindray BS-230), and findings may not generalize to other direct LDL-C assay systems used across Uganda or sub-Saharan Africa. Repeated lipid profiles from the same patients were treated as independent observations, which may have affected precision estimates, though this approach reflects real-world clinical practice where each test informs decisions independently. The subgroup with severe hypertriglyceridemia was small ($n = 33$), reducing statistical precision and limiting the strength of conclusions in this stratum. Finally, clinical outcomes were not assessed, and the impact of method differences on patient management could not be evaluated. Studies incorporating beta quantification, fasting status, multi-analyzer platforms, larger hypertriglyceridemia subgroups, and clinical outcome data would strengthen the evidence base for laboratory practice recommendations in Uganda.

Conclusion

Agreement and misclassification between directly measured LDL-C, Friedewald, and Martin/Hopkins equations relative to the Sampson equation varied by triglyceride level. All methods showed comparable concordance with Sampson equation in the low-TG stratum, while performance diverged substantially at moderately elevated triglycerides, where Friedewald equation showed the poorest agreement and highest misclassification. In severe hypertriglyceridemia, larger differences were observed between all methods and Sampson-calculated LDL-C values, though conclusions in this stratum are limited by the small sample size. These findings highlight the importance of transparent reporting of the LDL-C measurement method used, and careful interpretation of results near ESC/EAS treatment thresholds, particularly in patients with elevated triglycerides.

List of abbreviations

ASCVD: Atherosclerotic cardiovascular disease, CHE: Cholesterol esterase, CHO: Cholesterol oxidase, CLIA: Clinical Laboratory Improvement Amendments, CVD: Cardiovascular disease, dLDL-C: Directly measured low-density lipoprotein cholesterol, ESC/EAS: European Society of Cardiology/European Atherosclerosis Society, GLM: General linear model, HDL-C: High-density lipoprotein cholesterol, HIS: Hospital information system, LDL-C: Low-density lipoprotein cholesterol, MAD: Mean absolute difference, NHANES: National Health and Nutrition Examination Survey, POD: Peroxidase, RCF: Relative centrifugal force, RMSE: Root mean square error, TC: Total cholesterol, TEa: Total allowable error, TG: Triglycerides, TOOS: N-ethyl-N-(2-hydroxy-3-sulfopropyl)-m-toluidine sodium, VLDL-C: Very low-density lipoprotein cholesterol.

Author contributions

BJ: Conceptualization, methodology, investigation, data curation, formal analysis, visualization, project administration, writing-original draft, **MC:** methodology, investigation, data curation, writing – review and editing, **GN:** methodology, investigation, data curation, writing – review and editing, **NDB:** methodology, investigation, data curation, writing – review and editing, **NS:** methodology, investigation, data curation, writing – review and editing, **HRB:** methodology, investigation, writing – review and editing, **KCE:** methodology, investigation, writing – review and editing, **NP:** methodology, investigation, data curation, writing – review and editing, **FS:** methodology, investigation, writing – review and editing, **NPP:** methodology, investigation, data curation, writing – review and editing, **DA:** investigation, data curation, writing – review and editing, **DW:** data curation, writing – review and editing, **JK:** methodology, investigation, **ES:** methodology, investigation, writing – review and editing, supervision, **BN:** methodology, investigation, writing – review and editing, supervision, **JBM:** Conceptualization, methodology, data curation, investigation, writing – review and editing, supervision, **GM:** Conceptualization, methodology, data curation, investigation, writing – review and editing, supervision, visualization, validation, **OA:** Conceptualization, methodology, data curation, investigation, writing – review and editing, supervision, visualization, validation

Ethics approval and consent to participate

This study was approved by the ethics committee of Sebbi Hospital. As the study involved the use of historical data, the need for informed consent from patients was waived by the Sebbi Hospital Ethics Committee in accordance with the guidelines of the Uganda National Council for Science and Technology (UNCST) guidelines. According to these guidelines, informed consent is not required for studies in which the research project carries no more than minimal risk, that is, risk no greater than the risks encountered in normal daily life in a stable society. Using historical data, well-anonymized, reduces the risk of harm to patients. This retrospective study, conducted using medical records review complied with the Helsinki Declaration. Ethical standards were maintained to promote and safeguard the privacy and confidentiality of the participants, with careful consideration of the benefits, risks, and burdens.

Consent for publication

Not applicable.

Availability of data and materials

The data that support the findings of this study is available at Figshare [doi:10.6084/m9.figshare.31369081](https://doi.org/10.6084/m9.figshare.31369081).

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.sciaf.2026.e03373](https://doi.org/10.1016/j.sciaf.2026.e03373).

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